Medi-Share

Modernizing Medi-Share's Billing Platform

At Medi-Share, our core financial applications relied on an aging frontend tangled with a massive Redux state store, alongside legacy backend packages that made feature delivery slow and prone to unexpected bugs. Rather than attempting a high-risk, all-at-once rewrite, I led a phased migration plan to modernize the stack piece-by-piece.

On the backend, I migrated our data layer from Sequelize to Prisma, upgraded all supporting packages, swapped out Luxon for DayJS, and implemented message queues to handle heavy data jobs without blocking the API. On the frontend, I dropped Redux entirely, as standard React capabilities had evolved to handle local and shared state cleanly, and established contract-first API design, strict Zod and class-validator DTO runtime validation, and discriminated union return types. I also introduced virtualized Material UI components for heavy data rendering and reorganized our repositories into a unified pnpm workspace.

Impact: Completely eliminated unnecessary global state overhead, drastically reduced production billing defects, and cut our CI/CD build times by over 40%.